CSE-574 Planning/Learning Methods AI

Type-1 Project Proposal

Thicket-Aware Regime Switching for Verifier-Grounded LLM Planning

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Thesis

Under a fixed planning budget, an LLM planner should not always adapt in the same way. A small controller can learn when to perform verifier-grounded learning, when to search for nearby adapter-space specialists, and when to stop adapting and verify candidate plans.

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1. Motivation

Recent advances in LLM-based planning present a fragmented picture. Fine-tuned models achieve high in-domain plan validity but fail catastrophically on unseen domains. Belcamino et al. (2026) fine-tuned a 1.7B-parameter planner on 40,000 PDDL tuples from ten IPC domains and reached 82.9% valid plan rate in-domain, yet achieved 0% on two held-out domains. Symbol anonymization and compact serialization changes that preserve plan semantics entirely caused sharp performance drops, revealing that the model had learned surface-form patterns rather than planning structure.

This brittleness is not limited to fine-tuned small models. The PlanBench evaluations (Valmeekam et al., 2023; 2024) demonstrated that even frontier LLMs collapse on Mystery Blocksworld, an obfuscated variant of the standard Blocksworld domain that replaces all predicate and action names with semantically irrelevant strings. Standard AI planners are entirely unaffected by such renaming. The obfuscation gap is one of the strongest empirical signals that LLM planning success relies on approximate retrieval rather than structural reasoning (Kambhampati et al., 2024).

In parallel, two lines of work suggest alternative adaptation mechanisms. He et al. (2026) showed that unsupervised reinforcement learning with verifiable rewards can improve models briefly but often leads to confidence-correctness misalignment and eventual collapse, arguing that external reward environments are the more promising long-run path. Gan and Isola (2026) introduced Neural Thickets, demonstrating that large pretrained models contain dense neighborhoods of task-specific experts discoverable by random perturbation and selection a

method called RandOpt that matches or outperforms gradient-based fine-tuning on classification benchmarks.

Silver et al. (2024) provide further evidence that different domains demand fundamentally different strategies. Their generalized planning approach prompting GPT-4 to synthesize Python solve functions validated against training tasks achieved strong results on five of seven PDDL domains but failed on the remaining two for structurally distinct reasons. In Miconic, GPT-4 could not reliably infer that multiple buildings exist only implicitly through floor-ordering predicates. In Spanner, it consistently failed to detect that locations form a one-way chain, proposing strategies that assumed bidirectional movement. Under name obfuscation, performance collapsed almost entirely across all domains.

These findings converge on a gap that none of them fills alone. Verifier-grounded learning helps on some instances but quietly destroys transfer on others. Local perturbation search is cheap when the weight neighborhood is dense with specialists, but wasteful when it is not. Verified candidate selection is safe but leaves headroom where the model is almost correct. The missing formulation is a decision layer that allocates limited adaptation budget across these regimes based on instance-level signals. This proposal introduces that formulation.

The work sits within the LLM-Modulo framework (Kambhampati et al., 2024), which argues that LLMs should not be expected to plan or self-verify autonomously but can serve as approximate knowledge sources within tight verifier interaction loops. This proposal accepts that architecture and asks the next question: given a verifier-grounded loop with a fixed budget, how should the planner decide whether to learn, search, or verify?

2. Research Questions and Hypotheses

- **RQ1.** Can a learned controller allocate budget across learning, local search, and verification better than any fixed single-regime strategy on unseen-domain planning tasks?
- **RQ2.** Can cheap probes of trainability, thicket density, and generalization risk predict which regime should be used next?
- **RQ3.** Do nearby adapter-space specialists provide complementary gains to verifier-grounded learning for structured action generation?
- **RQ4.** Does structured aggregation outperform best-of-N selection enough to justify its complexity?

H1. Under matched budget, regime switching improves unseen-domain verified success over any fixed-regime baseline.

H2. Probes based on serialization sensitivity, anonymization sensitivity, and local perturbation gain are useful predictors of when further learning will overfit rather than generalize.

H3. When thicket density is high, adapter-space search is more valuable than another gradient update.

H4. Best-of-N validated selection captures most aggregation gain; richer methods are optional.

3. Problem Formulation

Let x denote a planning instance sampled from domain d. Let $M_{\theta 0}$ be a pretrained or lightly fine-tuned LLM planner. Let $V(x, p) \rightarrow \{0, 1\}$ be an external validator (VAL) returning whether candidate plan p is executable and goal-reaching for instance x. Define the per-instance adaptation budget as a triple:

$$B = (B_{tok}, B_{val}, B_{time})$$

where B_{tok} is the maximum tokens generated, B_{val} is the maximum VAL calls, and B_{time} is the maximum wall-clock time. All three components are fixed before experiments and held constant across all methods.

At adaptation round t, the controller observes a state summary s_t and selects an action a_t:

$$s_t = [q_t, d_t, r_t, b_t]$$

where q_t encodes trainability signals (verifier acceptance rate, prefix repair success, confidence-correctness alignment), d_t encodes thicket signals (perturbation gain distribution, candidate diversity), r_t encodes generalization-risk probes (anonymization sensitivity drop, compact serialization drop), and b_t encodes remaining budget. The controller π_φ selects from a discrete action set:

$$a_t = \pi_\varphi(a_t \mid s_t), \quad a_t \in \{LEARN, SEARCH, VERIFY, STOP\}$$

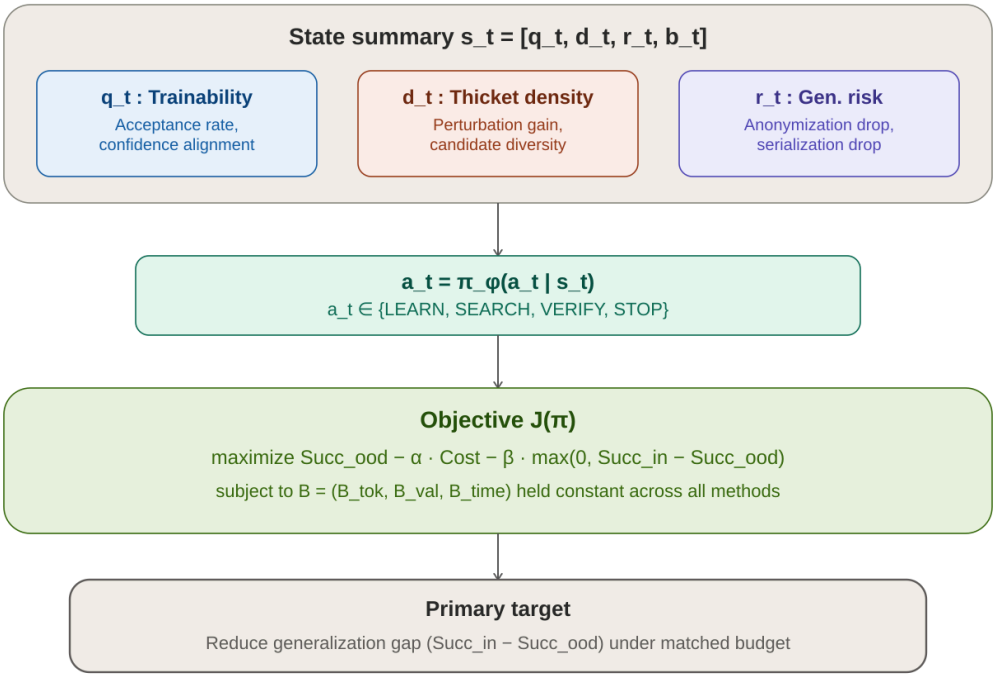
A regime choice for instance x at round t is **correct** if it is the action a_t that maximizes expected verified success on the held-out domain under the remaining budget b_t.

The objective function penalizes both cost and generalization failure:

$$J(\pi) = E[\, Succ_{ood} - \alpha \cdot Cost - \beta \cdot \max(0, Succ_{in} - Succ_{ood}) \,]$$

where $Succ_{ood}$ is the verified solved rate on held-out domains, Cost is the total adaptation cost under budget B, and the gap penalty explicitly discourages methods that improve in-domain validity at the expense of transfer.

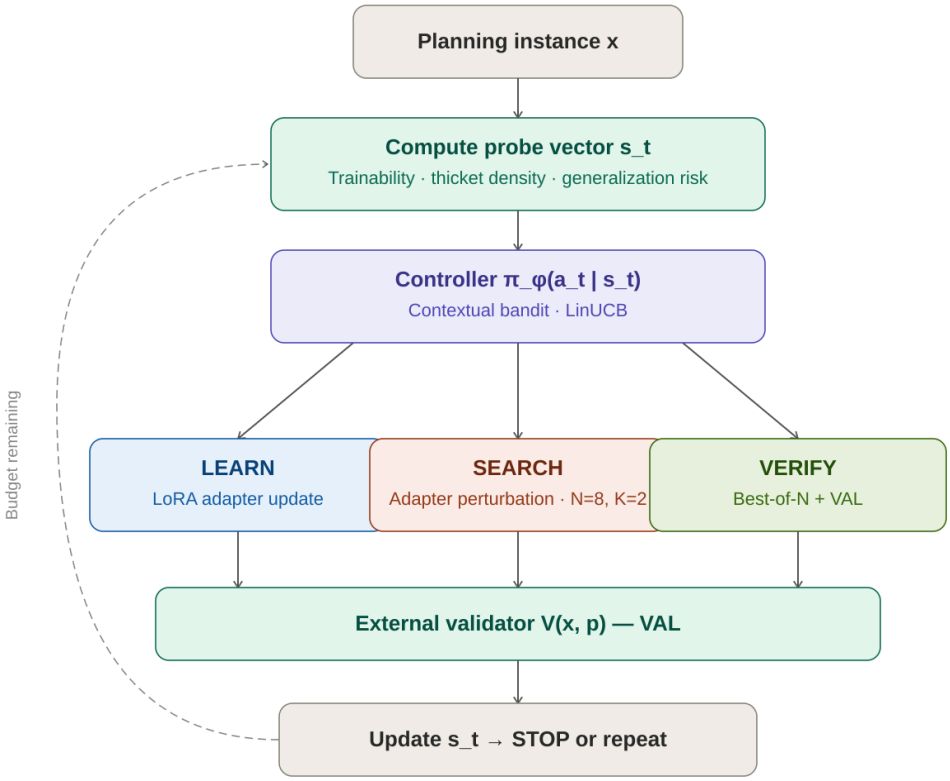
Figure 1. Problem formulation: budgeted regime allocation



4. Proposed Method

The system comprises five coupled modules: a base planner that generates candidate plans or action prefixes; a verifier that checks them via VAL; a learning module that performs tiny verifier-grounded adapter updates; a thicket-search module that samples nearby adapter-space specialists; and a regime controller that estimates the marginal value of each regime under remaining budget.

Figure 2. Regime switching controller: per-instance decision loop



4.1 Regime A: verifier-grounded learning (LEARN)

A single LoRA adapter update on a small support buffer of verified plan traces from training domains. The update is restricted to low-rank parameterization and limited to at most two rounds per instance. The round is abandoned if confidence increases without verified gains, following the collapse-risk finding from He et al. (2026).

4.2 Regime B: thicket search in adapter space (SEARCH)

Sample N = 8 Gaussian perturbations in adapter space across three noise scales ($\sigma \in \{0.001, 0.01, 0.1\}$). Evaluate each on four support instances via VAL. Retain the K = 2 best-performing specialists. The key diagnostic is thicket density: the fraction of perturbations that produce at least one additional valid plan relative to the base model. At most two search rounds.

4.3 Regime C: verified inference (VERIFY)

Generate multiple candidate plans from the current model state and score them with the external validator. The v1 system uses best-of-N validated selection with optional prefix-level repair. A verifier-scored prefix lattice is retained as an extension, activated only if best-of-N leaves measurable headroom.

4.4 Controller: probe features and decision rule

The controller is a contextual bandit (LinUCB with Thompson Sampling) receiving the probe vector s_t. Three probe families feed into the decision:

Table 1. Probe-to-regime decision guide

Probe signal	Example value	Favors	Disfavors
High anonymization drop	Accuracy drops >30%	VERIFY or SEARCH	LEARN
High serialization drop	Accuracy drops >20%	VERIFY	LEARN
High thicket density	≥4 of 8 perturbations yield new valid plans	SEARCH	LEARN
Low thicket density	0–1 of 8 improve	VERIFY or LEARN	SEARCH
Positive one-step gain	LoRA step improves verifier rate >5%	LEARN	STOP
High confidence, low validity	Model confident on invalid plans	VERIFY only	LEARN

Probe cost per instance: ~8 perturbation forward passes + 2 anonymized + 2 serialized + 1 gradient step + base passes. For a 1.7B model on one A100, approximately 30–45 seconds.

4.5 Controller training

Trained offline on meta-data from training domains only. For each instance: compute probe vector, run counterfactual one-step continuations for each regime, record delta in verified success. Fit a linear bandit on these tuples. Domain-family-level splits prevent controller memorization. Surface-form randomization applied during controller training.

5. Experimental Plan

5.1 Benchmarks and domain splits

Role	Domains	Rationale
Train (8)	Blocksworld, Gripper, Ferry, Delivery, Childsnack, Floortile, Rovers, Spanner	Diverse mix. Spanner retained because its one-way corridor is learnable with feedback but invisible under anonymization.
Held-out (4)	Miconic, Sokoban, Transport, Satellite	Structurally distinct. Miconic is particularly valuable: buildings exist only implicitly via floor-ordering (Silver et al., 2024).
Stress tests	All 12 domains × {standard, anonymized, compact-serialized}	Primary diagnostic for surface-form memorization. Motivated by PlanBench Mystery Blocksworld (Valmeekam et al., 2023).

Fresh instances generated using PDDL generators from AI-Planning/pddl-generators to mitigate data contamination.

5.2 Baselines

Baseline	Purpose	Required
Always LEARN	Verifier-grounded adaptation alone	Yes
Always SEARCH	Thicket-style local specialists alone	Yes
Always VERIFY	Generation + checking without adaptation	Yes
Hand-tuned schedule	Engineering baseline without learned controller	Yes
Rule-based controller	Threshold rules on probe features	Yes
Random controller	Sanity check	Yes
Oracle hindsight	Upper bound: best regime per instance post-hoc	Yes
LLM-as-gen.-planner	Silver et al. program synthesis on held-out domains	Yes
Prompted frontier LLM	GPT-5 zero-shot (standard + obfuscated)	Optional
Naive Cycle / Force-Mix Controller	Cycles through LEARN → SEARCH → VERIFY or enforces fixed proportions of each regime under identical budget, isolating gains from diversity alone	Yes

5.3 Metrics

- **Primary:** Verified solved rate on unseen domains under matched budget.
- **Generalization gap:** Succ_in − Succ_ood. The target is gap reduction, not raw accuracy.
- **Controller switching rate:** Percentage of instances where the controller selects a different regime than the best fixed baseline. Values below 15% indicate the controller is not doing meaningful work.
- **Efficiency:** Tokens generated, VAL calls, wall-clock time. All reported per instance.
- **Robustness:** Performance under anonymization and compact serialization stress tests.
- **Oracle alignment rate:** Fraction of rounds where the controller selects the same regime as the hindsight oracle**.**
- **Regret vs. hindsight oracle :** Difference in verified success between the controller's choices and the best possible sequence of regime choices under the same budget.
- **Diversity-controlled gain :** Improvement over diversity-matched baselines (random, cycle, force-mix), isolating the effect of regime selection from exploration.

5.4 Ablations

- Remove thicket-density probes from controller input.
- Remove generalization-risk probes (anonymization + serialization sensitivity).
- Remove trainability probes (one-step gain, confidence alignment).
- Budget ablation: small (2 rounds), medium (4 rounds), large (6 rounds).
- Domain ablation: block one family from controller training, test transfer.

5.5 Go / no-go criteria

- **Go** if the controller beats the best fixed regime by ≥8% absolute on held-out domains under matched budget.
- **Go** if gains survive both anonymization and serialization stress tests.
- **Go** if the controller switching rate exceeds 25% of instances.
- **No-go** if gains vanish after budget matching or appear only on training-like domains.
- **No-go** if removing thicket probes barely changes results. Reframe as generic budgeted allocation.

6. Related Work

Verifier-grounded learning for planning. Belcamino et al. (2026) integrated VAL as a reinforcement signal and showed partial recovery under compact serialization but no cross-domain improvement. He et al. (2026) demonstrated that RLVR can collapse when confidence misaligns with correctness. This proposal treats learning as one regime among several, activated only when probes indicate it is safe.

LLM-Modulo and limits of LLM planning. Kambhampati et al. (2024) argued that LLMs function as approximate knowledge sources incapable of autonomous planning or self-verification. The PlanBench evaluations (Valmeekam et al., 2023; 2024), including Mystery Blocksworld, established that obfuscation destroys planning performance. This proposal operationalizes the anonymization diagnostic as a runtime probe for the regime controller.

Generalized planning with LLMs. Silver et al. (2024) showed GPT-4 can synthesize generalized Python programs for PDDL domains, with strong results on structurally simple domains but consistent failure on Miconic and Spanner. Their No Names ablation confirmed heavy surface-form dependence. This proposal includes their approach as a baseline.

Neural Thickets. Gan and Isola (2026) showed pretrained models contain dense expert neighborhoods reachable by random perturbation. This proposal adapts the insight to planning: search occurs in adapter space, thicket density is a probe feature, and aggregation uses best-of-N validated selection rather than majority vote.

Algorithm selection in planning. Portfolio methods (IBaCop, Delfi, AutoFolio) select among pre-built planners using instance features. This proposal selects among adaptation regimes for a single LLM, using novel generalization-risk probes. The bandit formulation is related to GAMBLETA (Gagliolo & Schmidhuber, 2011).

7. Compute Resources

All experiments will run on ASU's Sol supercomputer: 56 GPU nodes, each with four NVIDIA A100 (80 GB), connected via 200 Gb/s HDR InfiniBand, backed by 4 PB BeeGFS scratch storage (Jennewein et al., PEARC 2023). The 1.7B base planner fits on a single A100. Estimated total: ~500 A100-hours (200h base reproduction, 50h probe computation, 100h controller meta-data, 150h full evaluation with ablations).

8. Execution Plan

Due to the tight deadline of the CSE-574 course project, we will complete the work through Phase 4 (rule-based controller evaluation and held-out testing). Phases 5 (contextual bandit training) and 6 (optional extensions) are explicitly designated as future work and will be pursued only if additional time becomes available after the course deadline.

Phase	Deliverable	Gate
1	Reproduce 1.7B baseline. Lock splits, budgets, protocol.	Stable reproduction; trustworthy compute accounting.
2	Implement LEARN, SEARCH, VERIFY + fixed-regime baselines + oracle.	At least one regime shows domain-dependent advantage.
3	Implement probes. Correlation study: do probes predict which regime helps?	At least two probe families show predictive value.
4	Train rule-based controller. Test on held-out domains.	Rule-based controller beats best fixed regime.
5	Train contextual bandit. Full ablations, stress tests.	Gains survive budget matching at ≥8% absolute.
6	Optional: prefix-lattice or code-generation appendix.	Only after main result is secure.

9. Expected Contributions

- A formal problem statement for budgeted regime allocation in LLM planning.
- A practical controller that learns when to learn, search, or verify.
- Concrete probe features with measured predictive value.
- A compute-matched evaluation protocol centered on held-out-domain success and anti-memorization stress tests.
- Evidence about when verifier-grounded learning helps planning and when it quietly hurts transfer.
- A clean negative-result path: if the controller fails, the paper clarifies which signals and regimes do not transfer.

10. Questions for Discussion

The following questions identify decision points where input would be most valuable before committing to implementation.

1.The proposed split trains on 8 IPC domains and holds out 4 (Miconic, Sokoban, Transport, Satellite). Should Spanner move to the held-out set given that its one-way corridor structure is also a known LLM blind spot?

2. The current design chooses one regime per round. An alternative is sub-round proportional allocation. Which should be the v1?

3. If adapter-space search contributes less than 3% absolute gain, should the paper still carry the 'thicket-aware' framing, or reframe as generic budgeted regime allocation?

4. The plan uses the 1.7B model from Belcamino et al. for direct comparability. Should we also run a 7B model from the start, or scale up only if the story holds at 1.7B?

5. Corrêa et al. (2025) showed GPT-5 matching LAMA on standard IPC tasks but degrading under obfuscation. Is including a prompted frontier LLM as a reference point worth the API cost?

6. If the probe correlation study shows strong predictive value but the learned controller adds only modest gains over the rule-based controller, is the probe analysis itself a publishable contribution? Should we consider a two-paper strategy?

7. PlanBench provides the obfuscation diagnostic that this proposal operationalizes as a runtime probe. Are there current or planned PlanBench extensions this work should coordinate with?

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